# Validation Pack

## LIMS validation evidence and assurance reports

Risk Assessment

Assurance Rationale

Test Scripts

Traceability Matrix

Gap Report

**DOCUMENT CONTROL**

| | | |
|---|---|---|
| **Project Name** | : | LIMS |
| **Version** | : | 1.0 |
| **Generated Date** | : | 29 Oct 2026 |
| **Generated By** | : | RAVEN System |
| **Classification** | : | Confidential |

## Table of Contents


| No. | Section | Page |
|---|---|---|
| 1 | Executive Summary | 3 |
| 2 | Risk Assessment | 4 |
| 3 | Assurance Rationale | 8 |
| 4 | Test Scripts | 12 |
| 5 | Traceability Matrix | 18 |
| 6 | Gap Report | 24 |
| 7 | Appendices | 28 |

# 1 Executive Summary

This validation pack provides a comprehensive overview of the risk assessment, assurance rationale, test scripts, traceability matrix and gap analysis for the LIMS system. The contents of this document support the validation activities and demonstrate compliance with applicable regulatory requirements.

### Key Information

| | | |
|---|---|---|
| Project | : | LIMS |
| Version | : | 1.0 |
| Generated Date | : | 29 Oct 2026 |
| Generated By | : | RAVEN System |
| Scope | : | Requirements, risks, test scripts and traceability for the LIMS module. |

## Validation Pack Components

1. **Risk Assessment**
2. **Assurance Rationale**
3. **Test Scripts**
4. **Traceability Matrix**
5. **Gap Report**

# 2 Risk Assessment

## 2.1 Risk Assessment Overview

The following table provides the identified risks for the requirements, along with their severity, probability, risk score and proposed mitigation.

| Req ID | Requirement | Risk | Severity | Probability | Risk Score | Mitigation |
|---|---|---|---|---|---|---|
| REQ-101 | System shall use Aptos Narrow font for display. | UI inconsistency | Low | Medium | 2 | Validate font configuration |
| REQ-102 | Electronic signatures shall be implemented. | Non-compliance | High | High | 9 | Enable Part 11 signature workflow |
| REQ-103 | Audit trail shall capture all changes. | Incomplete audit trail | Medium | Medium | 4 | Verify audit logging configuration |
| REQ-104 | Data shall be encrypted at rest. | Data breach | High | Medium | 6 | Enable encryption controls |
| REQ-105 | User access shall be role based. | Unauthorized access | Medium | Medium | 4 | Configure role based permissions |

# 3 Assurance Rationale

## 3.1 Rationale for Selected Requirements

Assurance decisions are supported by the recorded assessment, residual risk, confidence and rationale for each selected requirement.

**REQ-102** **Electronic signatures shall comply with FDA 21 CFR Part 11.**

| | |
|---|---|
| Assessment | Compliance controls detected |
| Residual Risk | High |
| Confidence | 96% |
| Assurance Statement | Compliance controls detected; human review completed; test coverage available. |

**REQ-103** **Audit trail shall capture all changes.**

| | |
|---|---|
| Assessment | Compliance controls detected |
| Residual Risk | Medium |
| Confidence | 88% |
| Assurance Statement | Requirement mapped to EU Annex 11; audit logging controls detected; test coverage available. |

**REQ-104** **Data shall be encrypted at rest.**

| | |
|---|---|
| Assessment | Compliance controls detected |
| Residual Risk | High |
| Confidence | 92% |
| Assurance Statement | Sensitive data controls identified; verify encryption configuration and access controls. |

**REQ-105** **User access shall be role based.**

| | |
|---|---|
| Assessment | Compliance controls detected |
| Residual Risk | Medium |
| Confidence | 90% |
| Assurance Statement | Role-based access controls identified; verify permission mapping and review process. |

# 4 Test Scripts

## 4.1 Test Script Details

### TC-001 Verify displayed font

**Objective**

Verify that the application uses Aptos Narrow font for display.

**Preconditions**

Application installed and user is logged in.

**Steps**

1. Open the dashboard
2. Navigate to Reports
3. Observe the displayed font.

**Expected Result**

The displayed font is Aptos Narrow.

**Pass Criteria**

All visible text uses Aptos Narrow.

### TC-002 Verify electronic signature workflow

**Objective**

Verify that electronic signatures are implemented as per requirements.

**Preconditions**

User has appropriate access.

**Steps**

1. Open a record
2. Initiate electronic signature
3. Complete the signature process
4. Verify the signed record.

**Expected Result**

The record is signed and stored successfully.

**Pass Criteria**

Signature is captured and audit trail is updated.

# 5 Traceability Matrix

## 5.1 Requirement Traceability

The following matrix shows the traceability between requirements, risks, regulatory references and test cases.

| Requirement ID | Risk ID | Regulatory Reference | Test Case ID | Status |
|---|---|---|---|---|
| REQ-101 | RSK-001 | - | TC-001 | Pass |
| REQ-102 | RSK-002 | FDA 21 CFR Part 11 | TC-002 | Pass |
| REQ-103 | RSK-003 | EU Annex 11 | TC-003 | Pass |
| REQ-104 | RSK-004 | - | TC-004 | Pending |
| REQ-105 | RSK-005 | - | TC-005 | Pass |
| REQ-106 | RSK-006 | EU Annex 11 | TC-006 | Pending |

# 6 Gap Report

## 6.1 Identified Gaps

The following table lists the gaps identified during the validation assessment, along with their impact and recommended actions.

| Requirement ID | Gap | Impact | Recommendation | Status |
|---|---|---|---|---|
| REQ-104 | No associated test case. | Medium | Generate test script. | Open |
| REQ-106 | Human review pending. | High | QA approval required. | Open |
| REQ-108 | No regulatory mapping. | Medium | Review applicability. | Open |
| REQ-112 | Incomplete test coverage. | High | Create additional test cases. | Open |
| REQ-115 | Missing evidence. | Low | Provide supporting evidence. | Open |